BEE Revision

Chapter-1

→ Branch and Nodes → Two or more branches meet
↓
Every circuit element constitute a branch

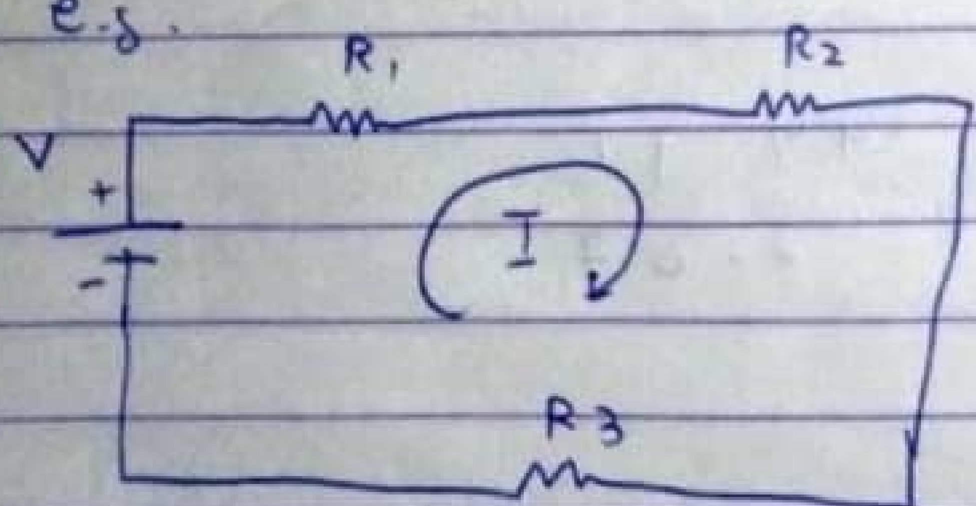
KCL:

Sum of current at node = 0
Incoming I = +ve
Outgoing I = -ve

KVL:

Sum of voltage rise / drop = 0
Voltage rise = -ve
Voltage drop = +ve

e.g.



$$\Rightarrow -V + IR_1 + IR_2 + IR_3 = 0$$

$$\Rightarrow I(R_1 + R_2 + R_3) = V$$

→ If something is not defined then we can't apply them.

Mesh analysis: (Loop)

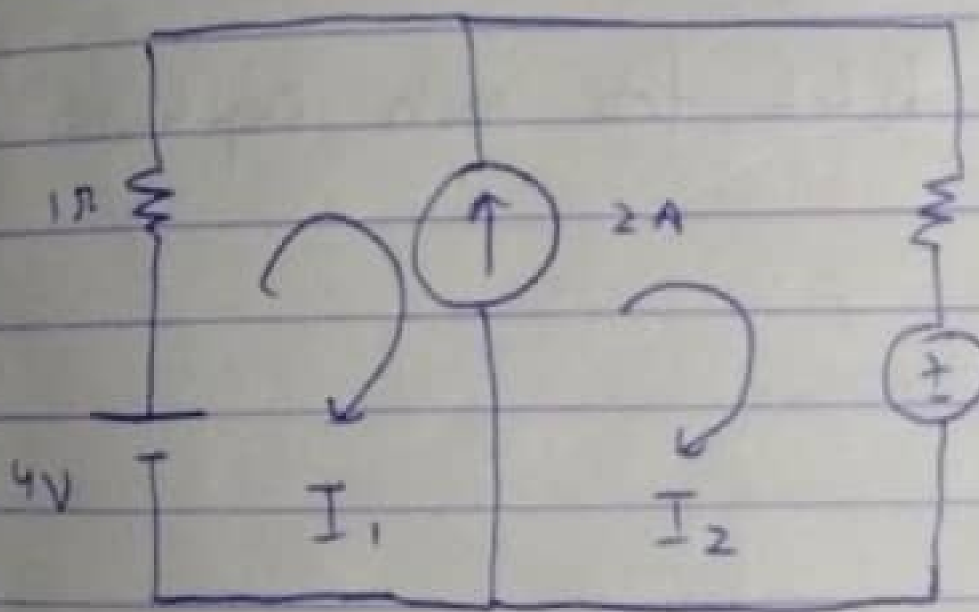
1) Identify total no. of loops and assign them loop current.
2) Apply KVL
3) Solve equations.

Total no. of equations = $b - n + 1$

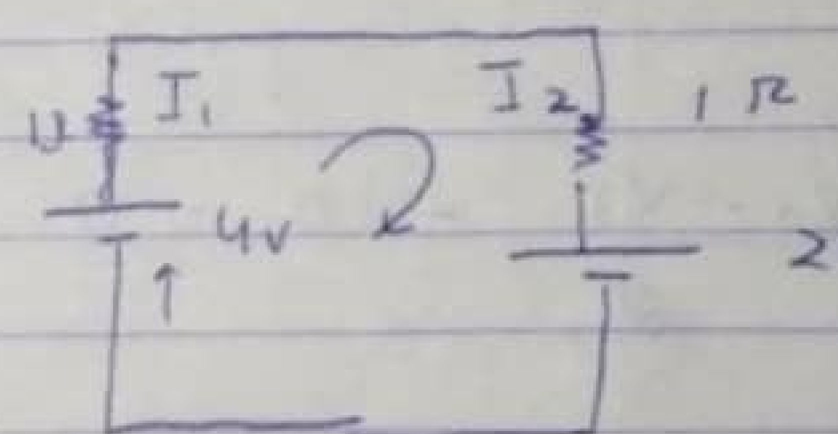
$b \longrightarrow$ No. of branches.

$n \longrightarrow$ No. of nodes.

Supermesh (Important):



This arrangement is super mesh.
(2A source b/w I_1 and I_2)
So, it is simplified as:



$\Rightarrow -4 + I_1 + I_2 + 2 = 0$

Also,

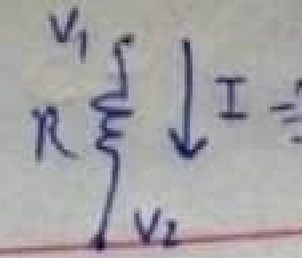
$2 = I_2 - I_1$ ($\rightarrow$ I_1 ↻ ↑2A ↺ I_2)

* Nodal analysis: → Identity total number of nodes (Only Principal)

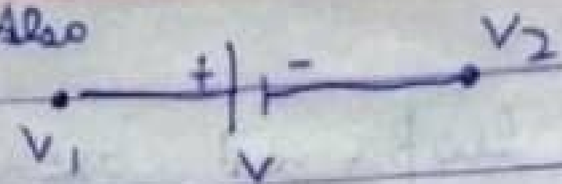
→ Take one node as reference node ($V = 0$)

→ Assign voltage to remaining nodes → Apply KCL → Total eqn: $(N-1)$

In doing $\frac{V_1 - V_2}{R}$, V_1 is generally initial voltage

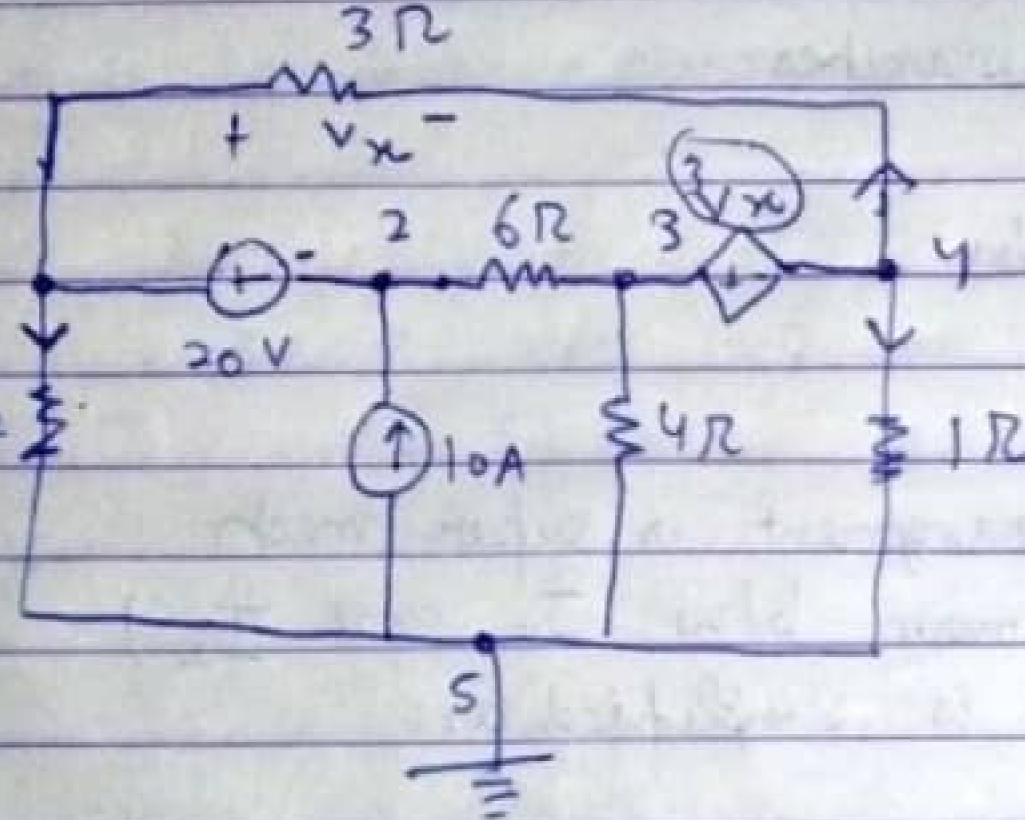
→ e.g 

$\Rightarrow \frac{V_1 - V_2}{R}$

Also 

$V = V_1 - V_2$

Super Node:

e.g. 

Total nodes = 5

1 and 2, 3 and 4 have ~~no~~ resistance in between so (1,2) and (3,4) constitute the supernodes

We will apply KCL for each supernode

For 1,2

$$\frac{V_1}{2} + \frac{V_1 - V_4}{3} + \frac{V_2 - V_3}{6} - 10 = 0$$

For 3,4:

$$\frac{V_3 - V_2}{6} + \frac{V_3}{4} + V_4 + \frac{V_4 - V_1}{3} = 0$$

Also,

$$\left.\begin{aligned} V_1 - V_2 &= 20 \\ V_3 - V_4 &= 3V_x \\ V_1 - V_4 &= V_x \end{aligned}\right\}$$

→ For finding equivalent resistance, identify the equivalent points and rearrange the circuit ~~for~~

* Star Delta :

Y – Δ

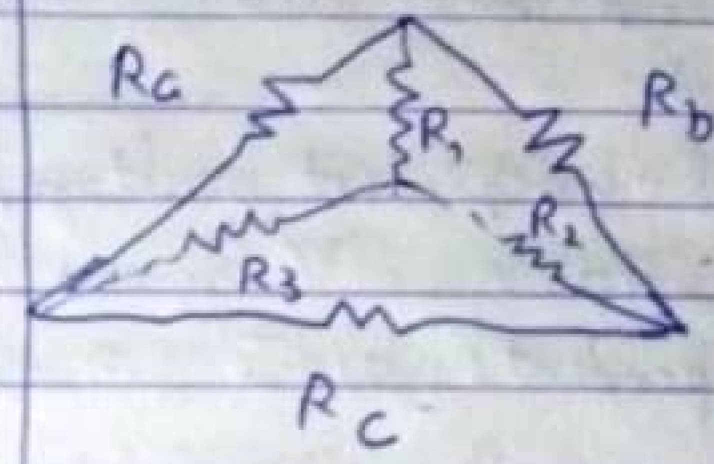
⇒ Star to Delta

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

→ Delta to Star :



$$R_1 = \frac{R_aR_b}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_bR_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_aR_c}{R_a + R_b + R_c}$$

* When resistors of equal value are transformed from Y – Δ

$$R_\Delta = 3R_Y$$

* When capacitor of equal —— Y to Δ

$$C_\Delta = \frac{1}{3} C_Y$$

(In capacitor same transformation can be done by doing $\frac{1}{C}$ instead of R in both LHS and RHS

S.C → Short Circuit , OC → Open Circuit

* Superposition Theorem:

In a linear, bilateral network containing more than one independent / dependent voltage / current source, the response in any element is the sum of the responses obtained with one source acting at a time and the other source being deactivated.

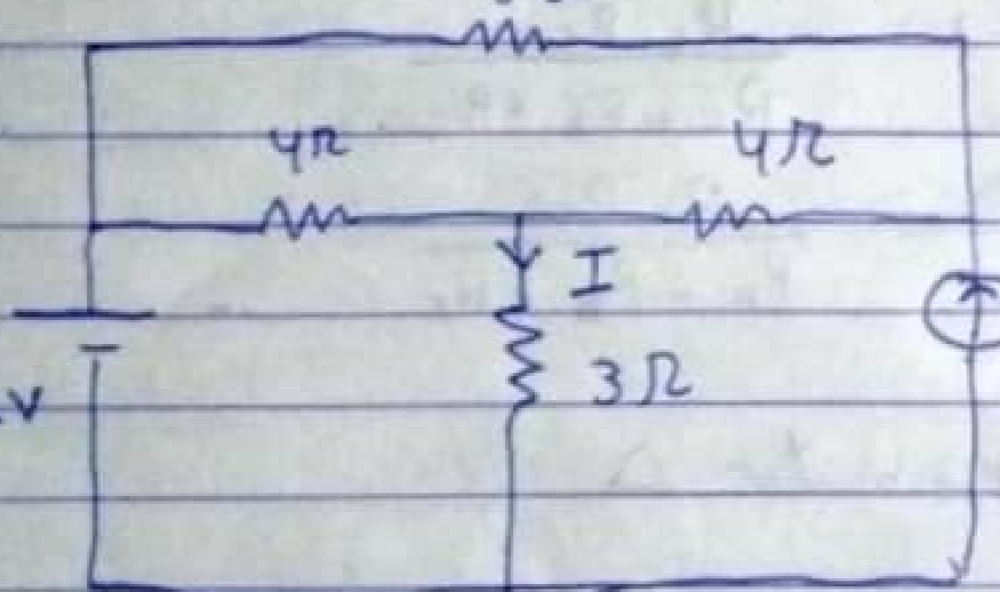
- → All independent voltage / current source are replaced by their internal resistance. V / Source → S.C , C / Source → O.C
- → Dependent source ⇒ No change

e.g

More than one source

Q Find I using superposition theorem

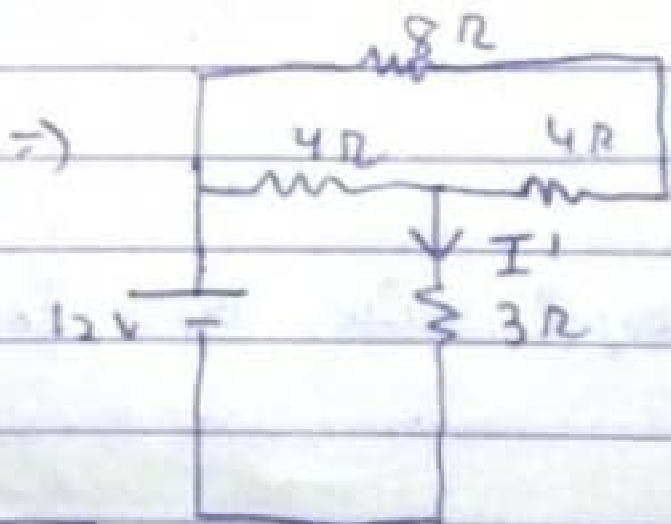
⇒



First consider 12V source

OC → Current Source

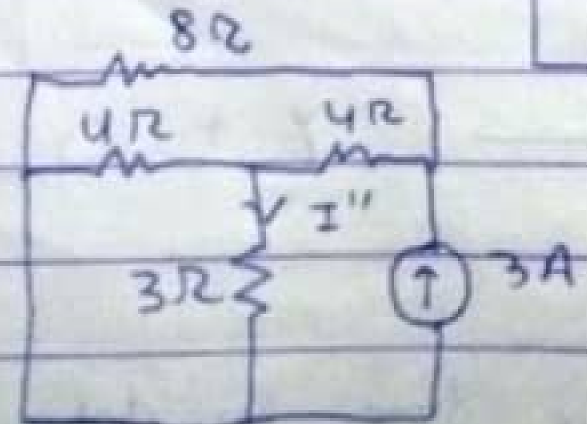
⇒



→ $I' = 2A$

Apply KCV

Now 3A source ⇒

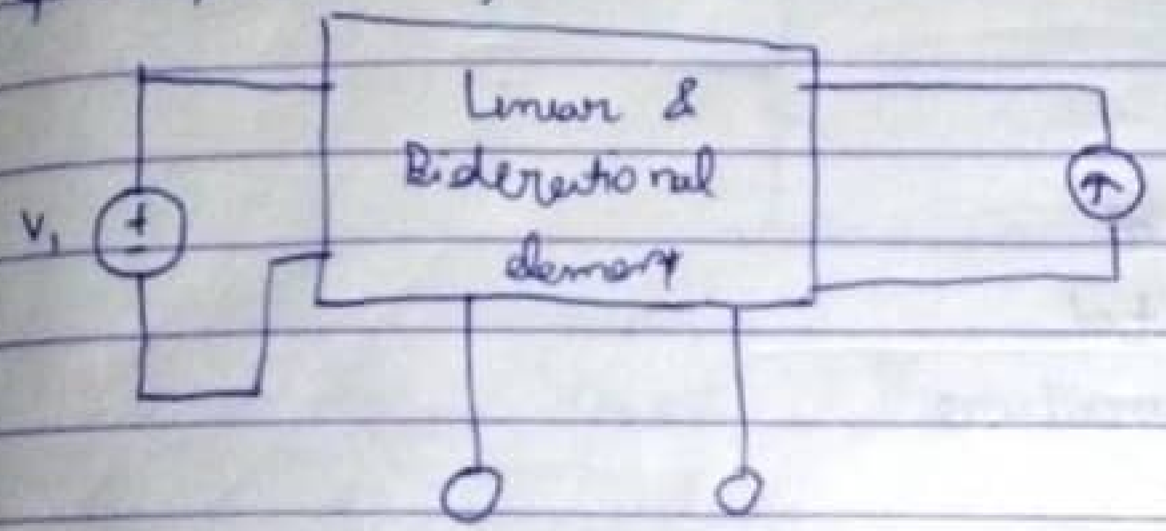


SC → VC

$I'' = 1A$

So, $I = I' + I'' = 1A + 2A = 3A$ Ans.

Q Important Question



V_1	I_1	V_0
5 V	0 A	3 V
0 V	2 A	6 V

Find V_0 when $V_1 = 10V$
$I_1 = 8A$

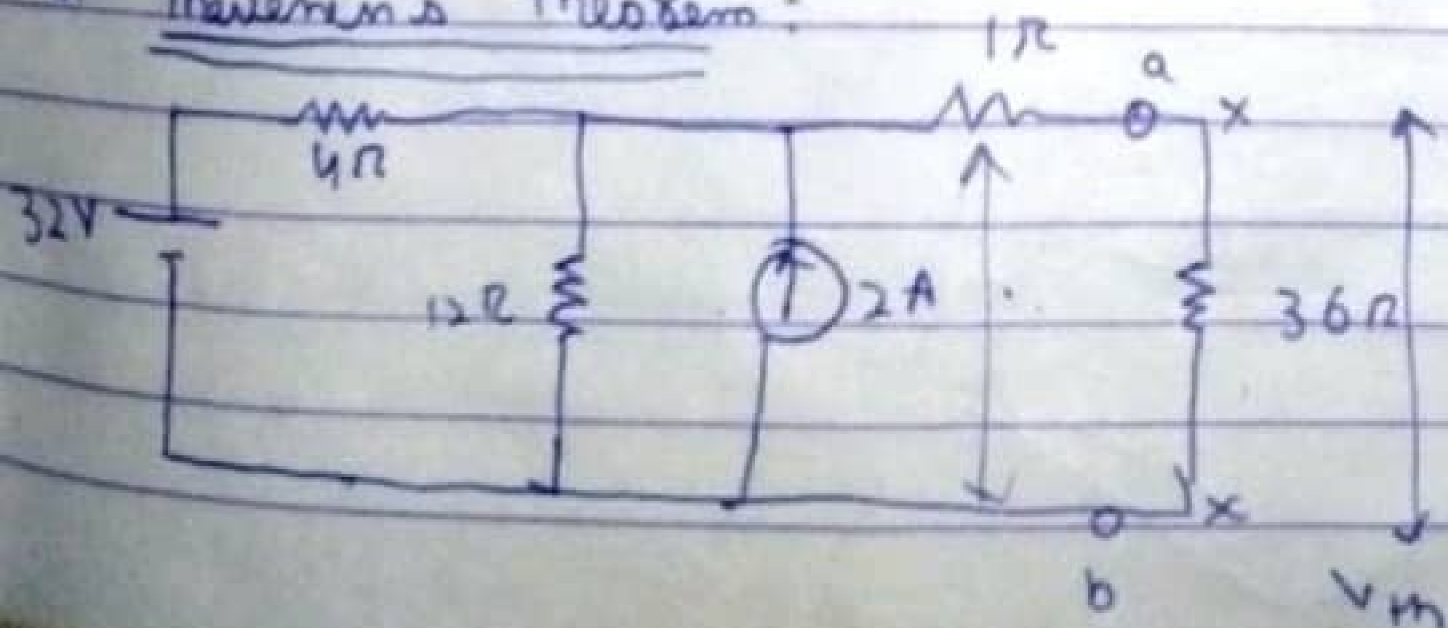
$\Rightarrow V_0 = \alpha V_1 + \beta I_1$

$\Rightarrow 3 = \alpha 5 + \beta(0) \Rightarrow \alpha = \frac{3}{5}$

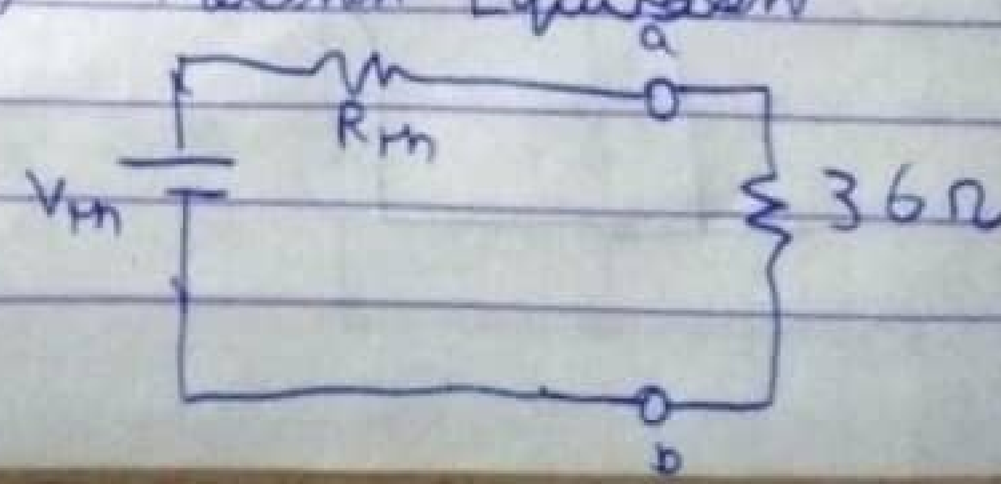
$6 = 2\beta \Rightarrow \beta = 3$

So, $V_0 = \cancel{10}^{2} \times \frac{3}{\cancel{5}} + 8 \times 3 = 6 + 24 = 30V$

* Thevenin's theorem:



$\Rightarrow$ Thevenin Equivalent



For V_{th}

Two terminal open circuited, Voltage across the open circuited terminal is V_{th}

For R_{th}

→ Terminal ab → Open

→ Independent Vtg source = Short circuited

⇒ " current " = Open circuited

→ Dependent Vtg / I source ⇒ Diff. procedure

Dependent Sources:

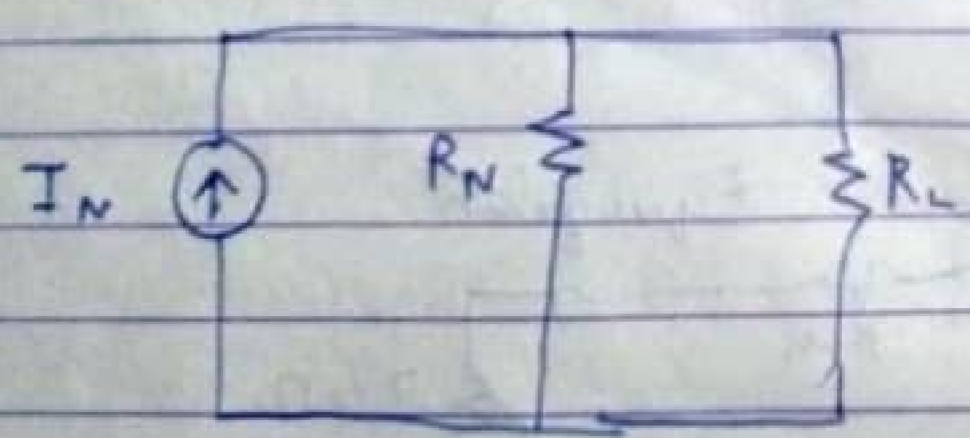
Three ways:

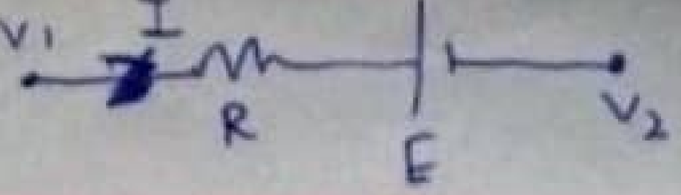
1) Connect 1V source across the terminal

2) Connect 1A source "

3) $R_{th} = \frac{V_{th}}{I_N}$

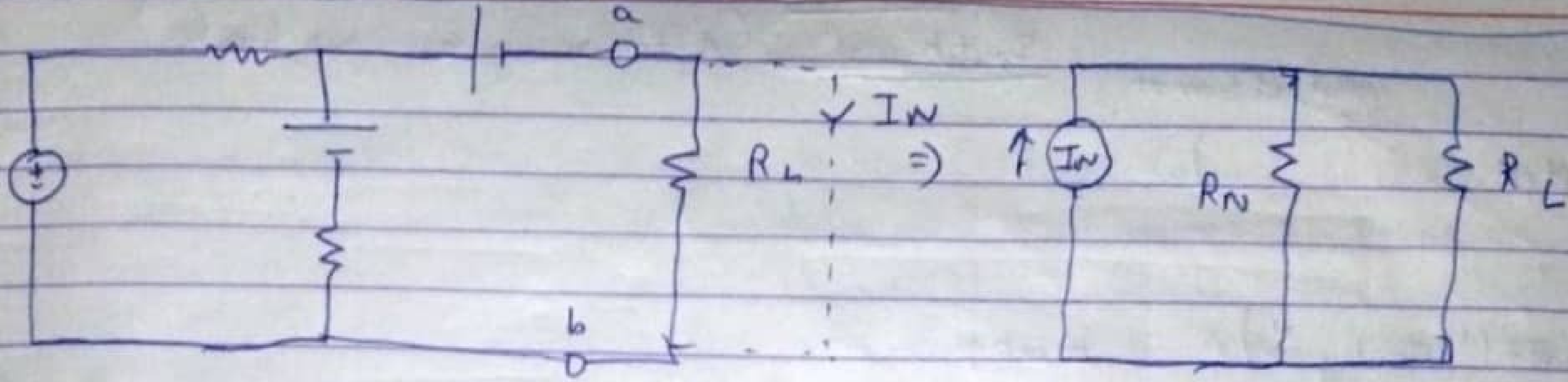
Norton Theorem:



* 

$\Rightarrow I = \left(\frac{V_1 - V_2 + E}{R} \right)$,, Reverse E polarity and it will be $-E$.

e.g.



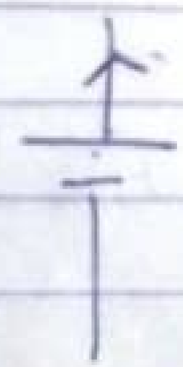
For I_N Short R_L

For R_N:
short all voltage source

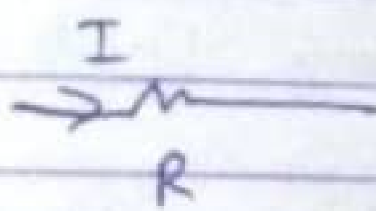
Maximum Power $\Rightarrow R_L = R_S$

Tellegen's $\Sigma P = 0$

P = +ve

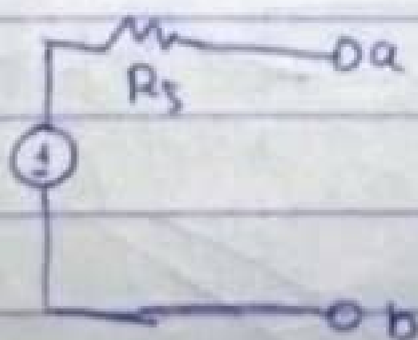


P = -ve

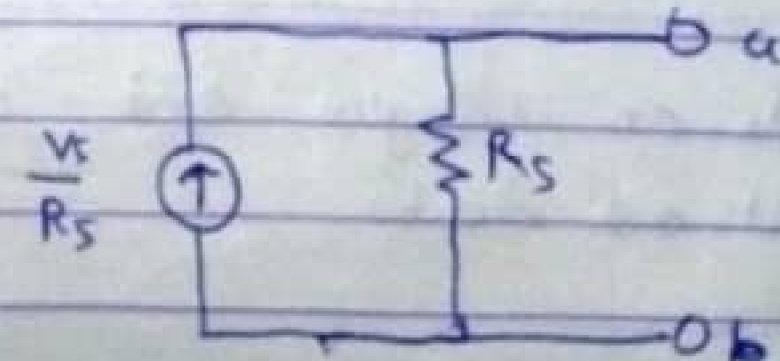


P = +ve

Source transformation:



$\Rightarrow$



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EE - 102

Chapter - 2

$\omega = 2\pi f$

$f = \frac{1}{T}$

x can be V or I whatever we need to calculate

$$\boxed{x_{av} = \frac{1}{T}\int_0^T x_m \sin t \, dt}$$

R.M.S value :

The voltage / current at which the heat dissipated in an AC circuit is equal to the heat dissipated in a DC circuit (provided both have same resistance and operated for same time).

$$\boxed{x_{r.m.s} = \sqrt{\frac{1}{T}\int_0^T (x_m \sin t)^2 \, dt}}$$

$$\boxed{\text{Form factor} = \frac{\text{Rms value}}{\text{Average value}}}$$

Peak is clearly the highest value reached

$$\boxed{\text{Crest factor} = \frac{\text{Max value}}{\text{R.M.S value}}}$$

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$v(t) = V_m \sin(\omega t + \phi) \Leftrightarrow V_m \angle \phi$ [Phasor]

While doing mathematical operations on phasors ensure the following:

- All the phasors must be a sine / cos func^n
- Same freq.
- Same sign of amplitude

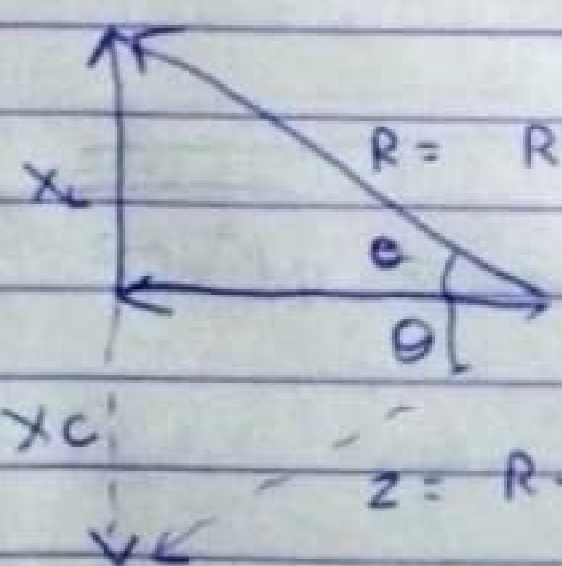
<u>Impedance</u>:

$$Z = \frac{V}{I} = R \pm jX \quad \text{or} \quad |Z| \angle \theta$$

Z is not a phasor

X_L or X_C (j just means 90° angle)

$$|Z| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1}\left(\frac{X}{R}\right) \quad \left(\begin{array}{l} R = |Z| \cos\theta \\ X = |Z| \sin\theta \end{array}\right)$$



$R = R + jX_L$

Impedance △

$Z = R - jX_C$

$$Y = \frac{1}{Z} = \frac{I}{V}$$ Admittance

$$G \pm jB = \frac{R}{R^2 + X^2} \pm \frac{jX}{R^2 + X^2}$$

G: Conductance, B: Susceptance

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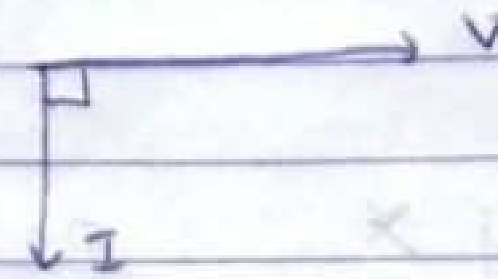
$X_L = j\omega L \rightarrow H$

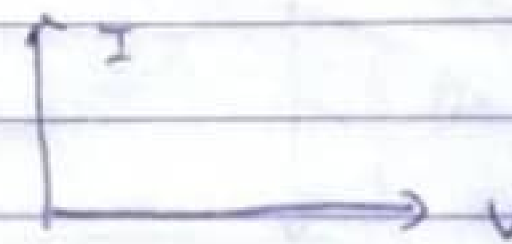
$X_C = \dfrac{1}{j\omega C} \rightarrow F$

$P_{avg} = V_{r.m.s} I_{r.m.s}$

* In inductor current lags by 90°



* In capacitor current leads by 90°



Complex Power $(S) = V_{r.m.s} I^{*}_{r.m.s}$

$S = P \pm jQ \rightarrow$ Reactive (var) unit

S → Complex Power (VA unit)

P → Real (W-unit)

$|S|$ is apparent Power $= \sqrt{P^2 + Q^2}$

$P = VI\cos\phi \qquad Q = VI\sin\phi$

$\boxed{S = P + jQ_L \text{ and } S = P - jQ_C}$

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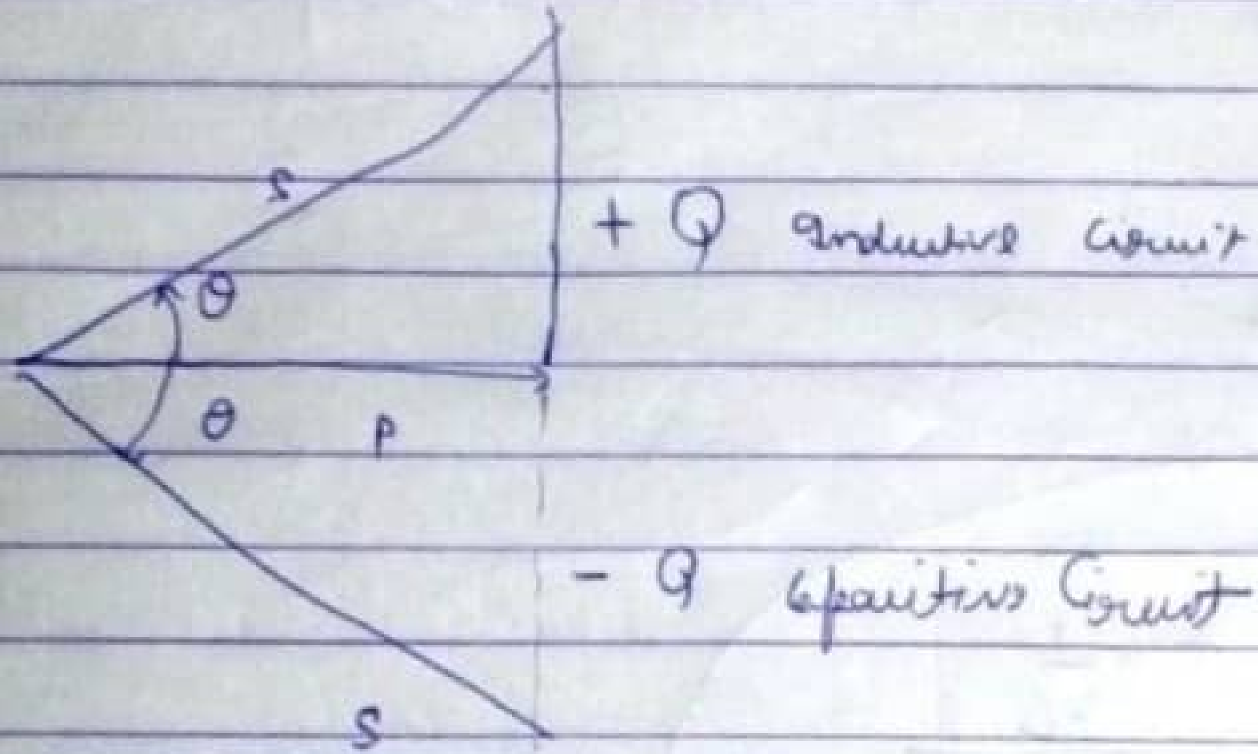
P
$\downarrow$
$I_{r.m.s}^2 R$ and $Q = I^2_{r.m.s} X$



$$\boxed{\cos\theta = \text{Power factor} = \frac{P}{S} = \cos(\theta_V - \theta_I)}$$

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}} , \quad f_0 = \frac{1}{2\pi\sqrt{LC}}} \rightarrow \text{Resonance}$$

~~Quality factor~~ $\cancel{Q_0 = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0 L}{R}}$

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Quality factor $Q_0 = \frac{w_0}{\Delta w} = \frac{\text{Max } \cancel{\text{power}} \text{ energy stored}}{\text{Energy dissipated}}$ } → per cycle

$Q_0 = \frac{1}{R}\sqrt{\frac{L}{C}}$ } series circuit

$= R\sqrt{\frac{C}{L}}$] → Parallel "

Lower Half Power freq : $\frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$

Upper ———— $= \frac{+R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$

Series

Just remember one thing

$w_1 = \text{lower} = w_0 - \frac{1}{2}\Delta w$

$w_2 = w_0 + \frac{1}{2}\Delta w$

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Parallel RLC :

$$\frac{1}{R} = \frac{1}{X_L} \qquad \frac{1}{X_C} \qquad \frac{1}{Z}$$

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